Proton Stability from Recognition Science: Color-Singlet Protection, No Grand Unification, and Exact Baryon-Number Conservation

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Abstract

We derive absolute proton stability from the Recognition Science (RS) framework. Three independent structural mechanisms protect the proton: (1) color confinement from the ledger's face-pair structure forbids decay to free quarks; (2) the gauge group $SU(3) \times SU(2) \times U(1)$ is derived from $D = 3$ cube geometry and does not unify into a larger group at any scale, so no X or Y bosons exist to mediate baryon-number violation; (3) baryon number is an exact ledger quantum number, with B-violating configurations incurring divergent J-cost. The prediction is $\tau_p = \infty$: the proton is absolutely stable. The current Super-Kamiokande bound $\tau(p \to e^+\pi^0) > 2.4 \times 10^{34}$ years is automatically satisfied. Any observed proton decay event would falsify RS.

1 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

Three independent structural features of RS guarantee proton stability: (1) color confinement (ledger color singlet requirement); (2) absence of gauge unification (gauge group $SU(3) \times SU(2) \times U(1)$ derived from $D = 3$ cube geometry, not descended from a GUT); and (3) exact baryon-number conservation (double-entry ledger).

2 Introduction

Proton stability is one of the sharpest discriminators between the Standard Model (SM) and its extensions in the context of Recognition Science [2]. In the SM, baryon number B is an accidental global symmetry—it is conserved classically but violated by non-perturbative electroweak processes (sphalerons) [4]. Grand Unified Theories (GUTs) predict proton decay as a signature of quark–lepton unification at high energies [3].

The experimental status:

Channel	**Experiment**	**τ bound (yr, 90% CL)**
$p \to e^+\pi^0$	Super-K [6]	$> 2.4 \times 10^{34}$
$p \to \bar{\nu}K^+$	Super-K	$> 5.9 \times 10^{33}$
$p \to \mu^+\pi^0$	Super-K	$> 1.6 \times 10^{34}$
$p \to e^+\eta$	Super-K	$> 4.2 \times 10^{33}$

These null results have already excluded minimal $SU(5)$ and constrain SUSY GUTs. RS predicts that all proton decay searches will continue to yield null results indefinitely: $\tau_p = \infty$.

3 Mechanism 1: Color Confinement

Definition 3.1 (Color Charge from Cube Geometry)**.** In RS, the $D = 3$ cube has 3 face-pairs ($N_c = F/2 = 2D/2 = D = 3$). Each face-pair defines a color degree of freedom, giving $SU(3)$ color. Color confinement is the requirement that all physical states are color singlets—balanced entries in the recognition ledger.

Lemma 3.2 (No Free Quarks)**.** *Free quarks carry unbalanced color charge, which corresponds to an unbalanced ledger entry with divergent J-cost. Physical states must be color singlets.*

Proposition 3.3 (Proton as Minimal Color-Singlet Baryon — Empirical Fact)**.** *The proton is the lightest color-singlet state with baryon number $B = 1$.*

Proof. The quark content of the proton is uud, forming a color singlet $\epsilon_{ijk}u^i u^j d^k$ with $B = 1$. The neutron (udd, $m_n = 939.565$ MeV) is heavier than the proton ($m_p = 938.272$ MeV) by 1.293 MeV. All other baryons (Λ, Σ, Ξ, Ω, etc.) are heavier still. The proton is therefore the unique lightest $B = 1$ color singlet. (This is an empirical fact; a zero-parameter RS derivation of $m_p < m_n$ from the φ-ladder is presented in the companion paper on the proton–electron mass ratio.) □

Corollary 3.4. *The proton cannot decay to free quarks (color confinement forbids it) and cannot decay to a lighter baryon (none exists). The only potential decay channels involve B-violation ($p \to$ mesons + leptons).*

4 Mechanism 2: No Grand Unification

Proposition 4.1 (Gauge Group Non-Unification — Structural Argument)**.** *The RS gauge group $SU(3) \times SU(2) \times U(1)$ does not embed into a simple Lie group at any energy scale.*

Proof. In RS, the gauge group factors are derived from the $D = 3$ cube geometry:

- $SU(3)$: From the 3 face-pairs of the cube.
- $SU(2)$: From the spinor structure of $SO(3)$ rotations of the cube.
- $U(1)$: From the overall recognition phase.

These factors are determined by the combinatorics of the cube, which is fixed and does not change with energy. There is no energy scale at which these factors merge into a single simple group.

In GUTs, the separate gauge factors are postulated to descend from a simple group (e.g., $SU(5)$, $SO(10)$, E_6) at $M_{\text{GUT}} \sim 10^{15}$–$10^{16}$ GeV. In RS, the gauge group is *derived from below* (from the cube), not *descended from above* (from a larger group). The concept of gauge coupling unification at a single point is therefore not applicable. □

Corollary 4.2 (No X and Y Bosons)**.** *Since the gauge group does not unify, the X and Y bosons of $SU(5)$ (which carry both color and electroweak quantum numbers and mediate B-violating interactions) do not exist. The dominant proton decay channel $p \to e^+\pi^0$ (mediated by X boson exchange) is absent.*

Remark 4.3. This also eliminates B-violation from dimension-6 operators of the form $\mathcal{O}_6 \sim qqq\ell/M_{\text{GUT}}^2$, since these operators arise precisely from integrating out the heavy X and Y bosons. In RS, no such heavy mediators exist, so these operators are identically zero.

Proposition 4.4 (Dimension-5 Operators Absent)**.** *In SUSY GUTs, dimension-5 operators $\mathcal{O}_5 \sim qqq\ell/M$ mediate proton decay through channels such as $p \to \bar{\nu}K^+$. These operators arise from colored Higgsino exchange. In RS, supersymmetry is not a fundamental symmetry (the φ-ladder mass spectrum has no supersymmetric partner structure), so these operators are also absent.*

5 Mechanism 3: Exact Baryon-Number Conservation

Definition 5.1 (Baryon Number as Ledger Quantum Number)**.** In RS, baryon number B is encoded in the ledger's double-entry structure. Each quark carries $B = 1/3$, and the total baryon number of a ledger configuration is the sum of the quark B-charges, balanced by the antiparticle entries.

Proposition 5.2 (Baryon Number Conservation in the Present Epoch — Structural Argument)**.** *In the present epoch (after the electroweak-scale defect density has dropped), the recognition ledger conserves baryon number at every tick.*

Proof. The double-entry structure requires that every recognition event creates a debit–credit pair with equal and opposite quantum numbers. A B-violating process would create an unbalanced ledger entry—a debit without a matching credit (or vice versa). The J-cost of an unbalanced entry diverges:

$$J(B_{\text{debit}}/B_{\text{credit}}) \to \infty \quad \text{when } B_{\text{debit}} \neq B_{\text{credit}}. \tag{1}$$

Therefore, B-violating processes are energetically forbidden—not merely suppressed by a large mass scale, but structurally impossible. □

Remark 5.3 (Contrast with the Standard Model). In the SM, B is an accidental symmetry. Non-perturbative effects (sphalerons) violate $B + L$ while conserving $B - L$, at a rate exponentially suppressed at zero temperature ($\Gamma \sim e^{-8\pi^2/g^2}$) but unsuppressed at the electroweak phase transition temperature [4, 5]. In RS, B is *not accidental*—it is an exact conservation law of the ledger, and sphalerons do not violate it.

5.1 Baryogenesis Compatibility

Remark 5.4 (Baryogenesis Compatibility). The conservation of B in the present epoch does not conflict with the observed matter–antimatter asymmetry. In the companion paper on baryogenesis, B-violating processes (defect recombination) occur during the earliest ledger epochs when the defect density is extremely high. As the defect density drops (the electroweak transition analog), the B-violation rate becomes exponentially suppressed, and baryon number is effectively conserved from that point onward. The proton's stability in the present epoch is therefore consistent with cosmological baryogenesis: B violation was active only during the first few epochs and is structurally suppressed in the low-defect regime.

6 The Triple Protection Theorem

Proposition 6.1 (Proton Stability — Structural Conclusion)**.** *The proton is absolutely stable:* $\tau_p = \infty$.

Proof. Any proton decay channel must satisfy:

(i) Conservation of color: The final state must be a color singlet. Since the proton is the lightest color singlet with $B = 1$ (Proposition 3.3), decay to a lighter $B = 1$ state is kinematically impossible.

(ii) Conservation of baryon number: B-violating channels ($p \to$ mesons + leptons) would require $\Delta B = 1$, which is forbidden by exact B conservation on the ledger (Theorem 5.2).

(iii) No mediating particle: Even if B violation were possible in principle, the absence of X/Y bosons (Corollary 4.2) and dimension-5 operators (Proposition 4.4) eliminates all known decay mechanisms.

Each mechanism independently forbids proton decay; their conjunction makes the stability absolute. □

7 Comparison with GUT Predictions

Model	τ_p (yr)	Dominant channel	Status
Minimal $SU(5)$	10^{29}–10^{31}	$p \to e^+\pi^0$	Excluded
$SO(10)$	10^{33}–10^{35}	$p \to e^+\pi^0$	Testable
SUSY $SU(5)$	10^{34}–10^{36}	$p \to \bar{\nu}K^+$	Testable
Flipped $SU(5)$	10^{35}–10^{37}	$p \to e^+\pi^0$	Testable
RS	∞	None	Consistent

Remark 7.1. Every proton decay experiment to date has yielded a null result. RS is the only framework that predicts this as a *structural necessity* rather than a consequence of a large but finite M_{GUT}.

8 Experimental Tests

1. **Hyper-Kamiokande** [7]: With $\sim 10\times$ the fiducial volume of Super-K, Hyper-K will reach sensitivity $\tau(p \to e^+\pi^0) \sim 10^{35}$ years by the mid-2030s. RS predicts no events.

2. **DUNE** [8]: The liquid argon TPC will provide complementary sensitivity to $p \to \bar{\nu}K^+$ and other SUSY-favored channels. RS predicts no events.

3. **JUNO**: Water-based detection with sensitivity to several decay channels. RS predicts no events.

4. **Next-generation detectors**: Any future detector reaching $\tau_p > 10^{36}$ years would definitively exclude most GUT models while remaining fully consistent with RS.

9 Related Predictions

Corollary 9.1 (No Magnetic Monopoles)**.** *GUTs predict magnetic monopoles as topological defects from the $G_{\text{GUT}} \to SU(3) \times SU(2) \times U(1)$ symmetry breaking. Since RS has no GUT symmetry breaking, no magnetic monopoles are predicted. The null results from MACRO [9] and IceCube are consistent with RS.*

Corollary 9.2 (No Leptoquarks)**.** *GUT-scale leptoquarks (which carry both color and lepton number) are absent in RS, since they arise from representations of the unified group. Collider searches for leptoquarks should yield null results above the current exclusion limits.*

10 Predictions and Falsifiers

1. $\tau_p = \infty$: No proton decay in any channel, at any sensitivity. A single confirmed proton decay event would falsify RS.

2. **No X/Y bosons**: No production of GUT-scale mediators at any collider energy.

3. **No magnetic monopoles**: Null results in all monopole searches.
4. **No neutron–antineutron oscillation**: Since B is exactly conserved, n–$\bar{n}$ oscillations ($\Delta B = 2$) are forbidden.
5. **No free quarks**: Color confinement is exact; no free quarks will be observed.

11 Conclusion

Proton stability in RS is absolute: three independent structural mechanisms—color confinement, absence of grand unification, and exact baryon-number conservation—each independently forbid proton decay. The prediction $\tau_p = \infty$ is sharply falsifiable: a single proton decay event would disprove RS. The ongoing null results from Super-Kamiokande, and the anticipated null results from Hyper-Kamiokande, DUNE, and JUNO, are predictions of the framework.

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